MATH 345 Skeleton Notes

Unit 3: Beyond linear maps: local linearization and first-order optimization

Section 3.3: Partial derivatives, gradients, Hessians, and Jacobian matrices

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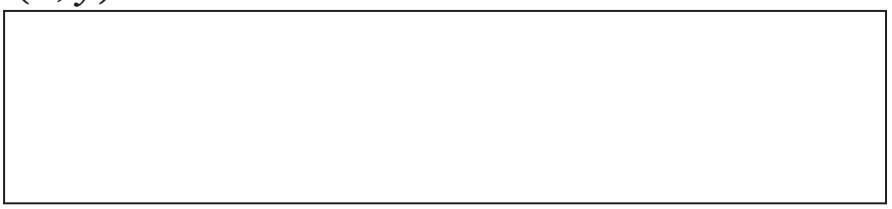
Partial derivatives

Definition and basic concepts

A partial derivative measures change in one input coordinate while the other input coordinates are held fixed.

Definition: Partial Derivatives

Let $f(x, y)$ be a scalar-valued function in two variables.

The  at (a, b) is the scalar quantity

$$f_x(a, b) = \lim_{h \to 0} \frac{f(a + h, b) - f(a, b)}{h},$$

provided this limit exists. Similarly,

the  at (a, b) is the quantity

$$f_y(a, b) = \lim_{h \to 0} \frac{f(a, b + h) - f(a, b)}{h}$$

provided this limit exists.

We compute the partial derivative in x of an expression precisely by treating the y-variable as a constant, and differentiating in x. Similarly, we compute the partial derivative in y by treating the x-variable as a constant.

Note

There are several notations used for the partial derivative of a function $f(x, y)$, namely, we might write $f_x(a, b)$ as

$$\partial_x f(a, b) \quad \frac{\partial f}{\partial x}(a, b) \quad (D_x f)(a, b) \quad \text{or} \quad (D_1 f)(a, b).$$

Similarly, we write $f_y(a, b)$ as

$$\partial_y f(a, b) \quad \frac{\partial f}{\partial y}(a, b) \quad (D_y f)(a, b) \quad \text{or} \quad (D_2 f)(a, b).$$

Example: Partial derivatives with one variable held constant (U3-LO3)

Let

$$f(x, y) = x^2 - 3xy + 2y^2 - 4x + 5y - 12.$$

Find $f_x(x, y)$ and $f_y(x, y)$ by differentiating with respect to one variable while keeping the other variable constant.

Activity: Computing first partial derivatives (U3-LO3)

For each function $f(x, y)$, find $f_x(x, y)$ and $f_y(x, y)$.
$f(x, y) = 3x^2y + 2x - 3y$

Activity: Computing first partial derivatives (U3-LO3)

For each function $f(x, y)$, find $f_x(x, y)$ and $f_y(x, y)$.
$f(x, y) = x \sin y + y \sin x$

Activity: Computing first partial derivatives (U3-LO3)

For each function $f(x, y)$, find $f_x(x, y)$ and $f_y(x, y)$.
$f(x, y) = x^2 e^{xy}$.

Activity: Reading first partial derivatives in SymPy (U3-LO3)

In Reading row reduction in code (U2-LO1, U2-LO2), SymPy kept matrix calculations exact. It can also differentiate symbolic expressions. The command `sp.symbols("x y")` creates two independent symbols, and `sp.diff(f, x)` differentiates `f` with respect to `x`, holding `y` fixed. Recall the function $f(x, y) = 3x^2y + 2x - 3y$ from our hand calculations.
In SymPy, `**` denotes exponentiation; for example, `x**2` represents x^2.

```
import sympy as sp

x, y = sp.symbols("x y")
f = 3*x**2*y + 2*x - 3*y
fx = sp.diff(f, x)
fy = sp.diff(f, y)

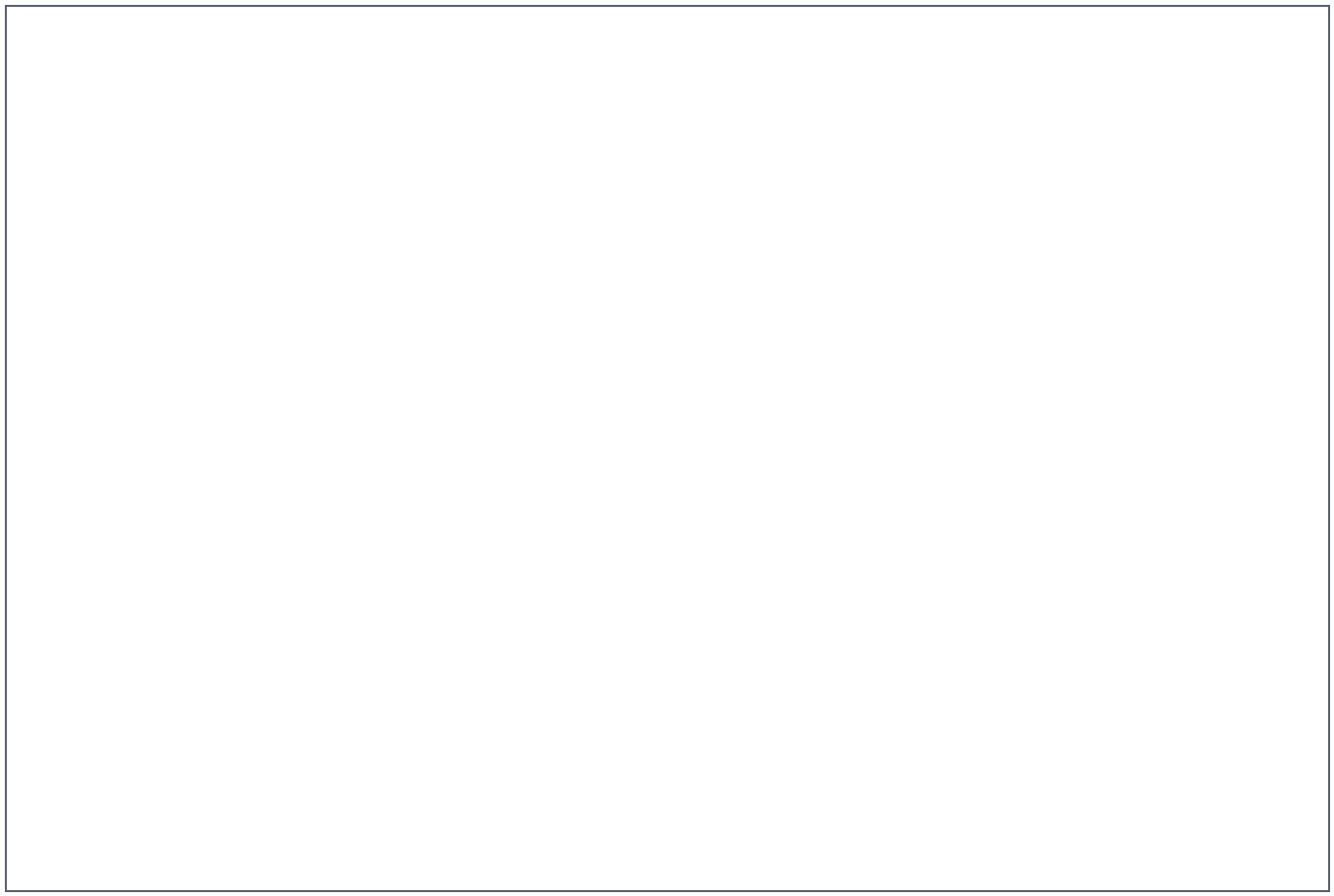
print((fx, fy))
print((fx.subs({x: 1, y: 2}), fy.subs({x: 1, y: 2})))
```

Output:

```
(6*x*y + 2, 3*x**2 - 3)
(14, 0)
```

The command `.subs({x: 1, y: 2})` substitutes $x = 1$ and $y = 2$ into an expression.

1. Match the two expressions in the first output line to f_x and f_y. Verify each by hand.
2. What is held fixed in each derivative? Interpret the two numbers in the second output line.
3. What would `sp.diff(f.subs({x: 1, y: 2}), x)` return? Why does this fail to compute $f_x(1, 2)$?



For the prediction loss in From predictions to a scalar error function (U3-LO1, U2-LO1), each input coordinate is a model weight. A partial derivative will measure how the total squared error changes when we adjust one weight while keeping the other weight and the observed data fixed.

Geometric interpretation

Definition: The Slice of a Function

Given a function $f(x, y)$ and $b \in \mathbb{R}$, the [ ] of the function f in the plane $y = b$ is the function $g : \mathbb{R} \to \mathbb{R}$ given by $g(x) = f(x, b)$. Similarly, for $x \in \mathbb{R}$ the slice of f in the plane $x = a$ is the function $g : \mathbb{R} \to \mathbb{R}$ given by $g(y) = f(a, y)$.

Geometrically, the slice of a function f in the plane $y = b$ corresponding to intersecting the surface given by the equation $z = f(x, y)$ with the

plane $y = b$, creating a curve in this plane.

Example: Sketching slices of a surface (U3-LO1)

Sketch the slices of the function $f(x, y) = \sin(x + y^2)$ in the plane $y = b$.

Theorem: Geometric Interpretation

If we consider the surface in $\mathbb{R}^3$ described by the equation $z = f(x, y)$, then

- $f_x(a, b)$ is the *slope of the tangent line* to the curve formed by intersecting the surface $z = f(x, y)$ with the vertical plane $y = b$.
- $f_y(a, b)$ is the *slope of the tangent line* to the curve formed by intersecting the surface $z = f(x, y)$ with the vertical plane $x = a$.

Activity: Partial derivatives as slopes (U3-LO3)

Let $f(x, y) = 16 - x^2 - y^2$. Find $f_x(1, 2)$ and $f_y(1, 2)$ and interpret these numbers as slopes of a certain curve.

Activity: Reading partial derivatives along a horizontal line (U3-LO3)

Consider the contour plot shown in the figure for a function F with contour lines at increments of 0.2. In particular, consider the curve obtained from the slice $y = 0$, indicated by the red line in the figure. By interpreting the contour plot, what can we say about the slopes of this curve, or equivalently, the values of F_x on this line?

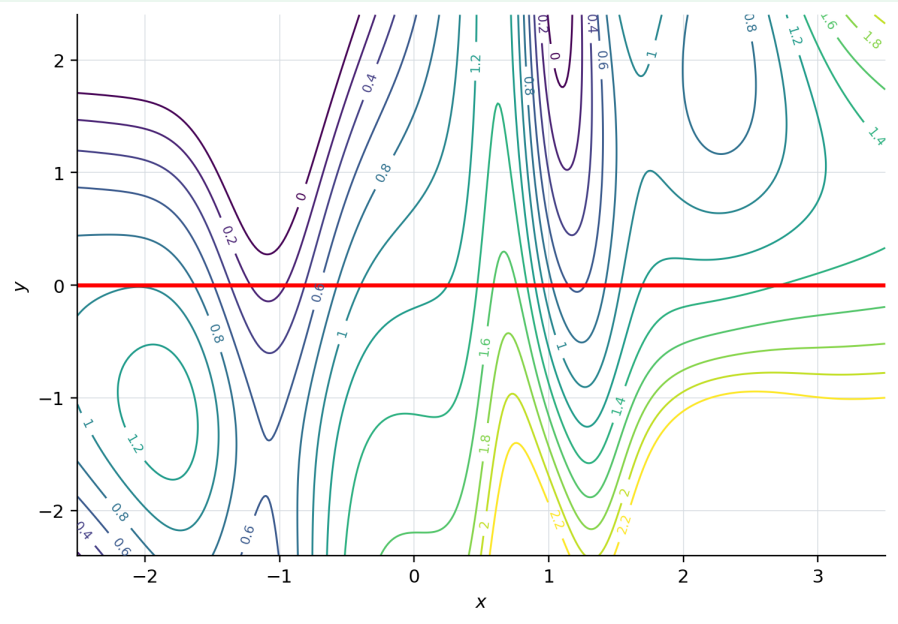


Figure. A contour plot for a function with increments of 0.2. The line $y = 0$ is indicated in red. Adapted from: Stanford's MATH 51 textbook

Activity: Reading partial derivatives along a vertical line (U3-LO3)

Consider the figure, which is the same contour plot as in the figure, but with the vertical line $x = 2.25$ indicated instead. As in Reading partial derivatives along a horizontal line (U3-LO3), read off the behaviour of the partial derivatives F_y from the contour plot.

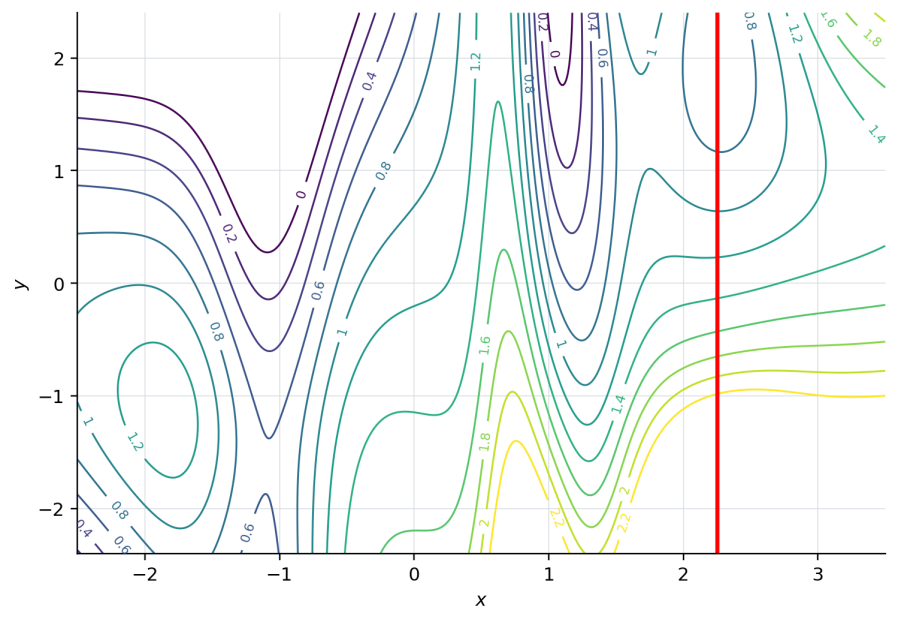


Figure. A contour plot with increments of 0.2, with the vertical line $x = 2.25$ highlighted in red. Adapted from: Stanford's MATH 51 textbook

Generally, the spacing of contour lines corresponds to the magnitude of partial derivatives, and the sign of the partial derivatives corresponds to whether the function increases or decreasing in that direction.

Activity: Checking derivative signs against a contour plot (U3-LO3)

Consider the function $f(x, y) = x(y^2 + 1)$, whose contour plot is indicated in the figure. Five points are marked on the plot: $P = (0, 0)$, $Q = (3/2, 0)$, $R = (7/2, 0)$, $S = (2, 1)$, and $T = (2, -1)$. For each of these five points, determine from the contour plot whether f_x is positive, negative, or zero, and whether f_y is positive, negative, or zero. Then calculate these derivatives exactly and check whether your observations were correct.

Activity: Checking derivative signs against a contour plot (U3-LO3) (continued)

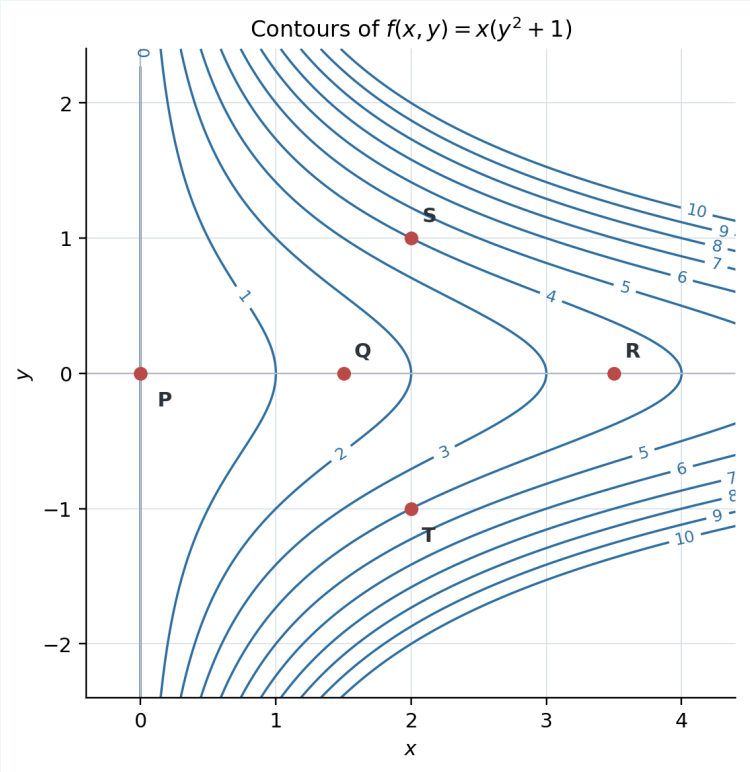


Figure. Contour plot for $f(x, y) = x(y^2 + 1)$ with increments of 1, and five points P, Q, R, S, and T indicated. Adapted from: Stanford's MATH 51 textbook

Partial derivatives with more variables

The concept of partial derivatives extends naturally to functions of more than two variables.

Definition

For a function $f(\mathbf{x}) = f(x_1, \dots, x_n)$ of n variables, the partial derivative with respect to x_i is defined as

$$f_{x_i}(\mathbf{x}) = \lim_{h \to 0} \frac{f(x_1, \dots, x_i + h, \dots, x_n) - f(x_1, \dots, x_n)}{h}$$

provided this limit exists. Using vector notation, we can write this definition more cleanly, by writing

$$f_{x_i}(\mathbf{x}) = \lim_{h \to 0} \frac{f(\mathbf{x} + h\mathbf{e}_i) - f(\mathbf{x})}{h},$$

where $\mathbf{e}_1, \dots, \mathbf{e}_n$ are the standard basis vectors (recall Standard basis). This partial derivative is also written as

$$\frac{\partial f}{\partial x_i} \quad \partial_{x_i} f \quad \text{or} \quad (D_i f)(\mathbf{x}).$$

If the variables $x_1, \dots, x_n$ are denoted using different symbols, these symbols might also be used to denote the partial derivatives instead. For instance, in $\mathbb{R}^3$, we might write a function as $f(x, y, z)$, and then the three partial derivatives of f may be denoted using any of the following four notations:

- $\frac{\partial f}{\partial x}$, $\frac{\partial f}{\partial y}$, and $\frac{\partial f}{\partial z}$.
- f_x, f_y, and f_z.
- $\partial_x f$, $\partial_y f$ and $\partial_z f$.
- $D_1 f$, $D_2 f$, and $D_3 f$.

As for functions of two variables, we can calculate these derivatives by treating all variables but that we are taking the partial derivative of as constants.

Activity: Partial derivatives of a product of powers (U3-LO3)

If $f(x, y, z) = 5x^2y^2z^3$, find f_x, f_y, and f_z.

Activity: Partial derivatives of a three-variable polynomial (U3-LO3)

Find f_x, f_y, and f_z where $f(x, y, z) = xy + yz + xz$.

Activity: Partial derivatives with logarithms and exponentials (U3-LO3)

Find f_x and f_y where $z = f(x, y) = x \ln y + ye^x$.

Higher-order partial derivatives

Given a scalar-valued function f, the partial derivatives of f (where defined) are also scalar-valued functions, so we can consider partial derivatives of that function as well, which leads to *higher order partial derivatives*.

Definition

If f is a function of two variables, then

the [ ] of f are defined as:

- The derivative f_{xx} or $\frac{\partial^2 f}{\partial x^2}$, obtained by taking the partial derivative of f twice in the x variable.
- The derivative $f_{xy} = (f_x)_y$ or $\frac{\partial^2 f}{\partial y \partial x}$, obtained by *first* differentiating f in the x variable, and *then* differentiating f in the y variable.
- The derivative $f_{yx} = (f_y)_x$, or $\frac{\partial^2 f}{\partial x \partial y}$, obtained by *first* differentiating f in the y variable, and *then* differentiating f in the x variable.
- The derivative f_{yy}, or $\frac{\partial^2 f}{\partial y^2}$, obtained by differentiating f twice in the y variable.

More generally, for a function $f(x_1, \ldots, x_n)$, the second-order partial derivatives are

$$f_{x_i x_j} = (f_{x_i})_{x_j} = \frac{\partial^2 f}{\partial x_j \partial x_i}$$

for $1 \leq i, j \leq n$, obtained by first differentiating f in the x_i variable, and then in the x_j variable. In operator notation, $f_{x_i x_j} = D_j D_i f$. Subscripts record the differentiation order from left to right; differential operators act from right to left.

Theorem: Clairaut's Theorem

Suppose a scalar-valued function f is defined in an open ball B that contains a point $\mathbf{a}$. If the functions $f_{x_i x_j}$ and $f_{x_j x_i}$ are well-defined and continuous on the ball B, then $f_{x_i x_j}(\mathbf{a}) = f_{x_j x_i}(\mathbf{a})$.

Activity: First and second partial derivatives (U3-LO3)

In this activity we compute all four partial derivatives of the function $f(x, y) = x \ln y + ye^x$.
First compute f_x and f_y.

Activity: First and second partial derivatives (U3-LO3)

In this activity we compute all four partial derivatives of the function $f(x, y) = x \ln y + ye^x$.
Next, compute f_{xx} and f_{yy}.

Activity: First and second partial derivatives (U3-LO3)

In this activity we compute all four partial derivatives of the function $f(x, y) = x \ln y + ye^x$.
Now compute f_{xy} and f_{yx}.

Note

For most functions f encountered in practice, the second-order partial derivatives are continuous where defined, and so Clairaut's Theorem applies. However, there are functions for which mixed partial derivatives are not equal to one another.

Activity: Second partial derivatives of a logarithm (U3-LO3)

Find all four second partial derivatives of the function

$$f(x, y) = \ln(x^3 + y^2).$$

Activity: Reading repeated and mixed derivatives in SymPy (U3-LO3)

Return to the polynomial in Reading first partial derivatives in SymPy (U3-LO3). The command `sp.diff(f, x, 2)` differentiates twice with respect to `x`. The command `sp.diff(f, x, y)` differentiates first with respect to `x`, then with respect to `y`; equivalently, it computes `sp.diff(sp.diff(f, x), y)`.

Activity: Reading repeated and mixed derivatives in SymPy (U3-LO3) (continued)

```
import sympy as sp

x, y = sp.symbols("x y")
f = 3*x**2*y + 2*x - 3*y
dxx = sp.diff(f, x, 2)
d_x_then_y = sp.diff(f, x, y)
d_y_then_x = sp.diff(f, y, x)
dyy = sp.diff(f, y, 2)

print((dxx, d_x_then_y, d_y_then_x, dyy))
```

Output:

```
(6*y, 6*x, 6*x, 0)
```

1. Match each output to f_{xx}, f_{xy}, f_{yx}, or f_{yy}, using this section's differentiation-order convention.
2. Starting from $f_x = 6xy + 2$ and $f_y = 3x^2 - 3$, verify the four outputs by hand.
3. Why does Clairaut's Theorem apply to this function? Does the matching output establish equality of mixed partial derivatives for every function?

Gradients and Hessians

The gradient collects first partial derivatives of a scalar-valued function.

Definition: Gradient

Let f be a scalar-valued function of n variables. At a point $\mathbf{x}$ where all first partial derivatives of f exist, the [ ] of f is the vector

$$(\nabla f)(\mathbf{x}) = \begin{bmatrix} \dfrac{\partial f}{\partial x_1}(\mathbf{x}) \\ \vdots \\ \dfrac{\partial f}{\partial x_n}(\mathbf{x}) \end{bmatrix}.$$

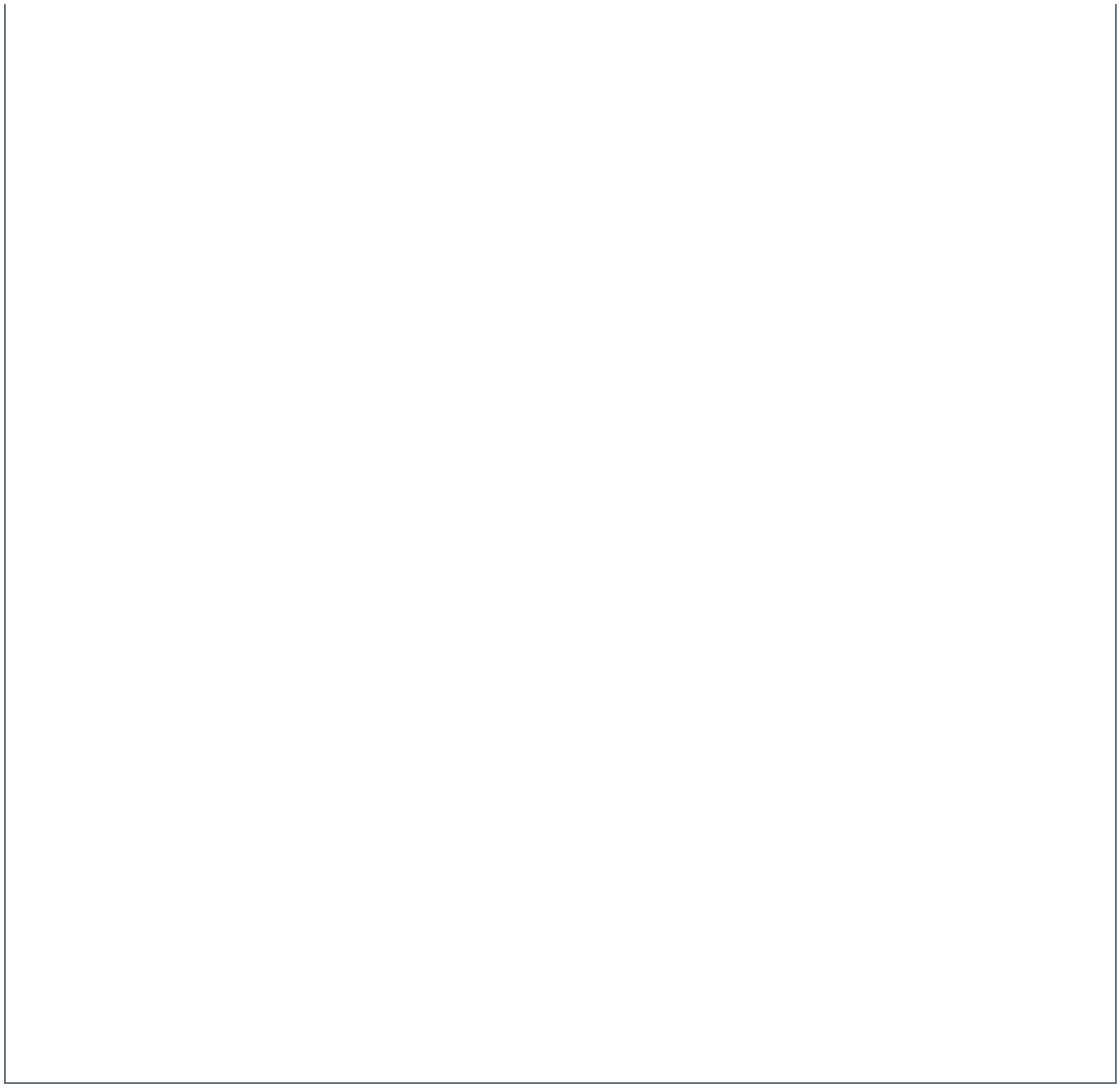
When these vectors are defined throughout a domain, they form a [ ] ∇f.

Activity: The gradient of a trigonometric function (U3-LO3)

Let $f(x, y) = \sin(x + y^2)$. Find the gradient of f.

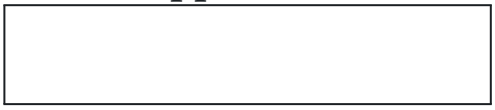
Activity: Evaluating a gradient at a point (U3-LO3)

Let $f(x, y) = x^2 + xy + y^2$. Compute $\nabla f(1, 2)$.



The Hessian collects second partial derivatives.

Definition

Let f be a scalar-valued function of n variables, and suppose the second partial derivatives of f exist at a point $\mathbf{x}$. The  of f at $\mathbf{x}$ is the $n \times n$ matrix of second partial derivatives:

$$H_f(\mathbf{x}) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2}(\mathbf{x}) & \frac{\partial^2 f}{\partial x_2 \partial x_1}(\mathbf{x}) & \cdots & \frac{\partial^2 f}{\partial x_n \partial x_1}(\mathbf{x}) \\ \frac{\partial^2 f}{\partial x_1 \partial x_2}(\mathbf{x}) & \frac{\partial^2 f}{\partial x_2^2}(\mathbf{x}) & \cdots & \frac{\partial^2 f}{\partial x_n \partial x_2}(\mathbf{x}) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_1 \partial x_n}(\mathbf{x}) & \frac{\partial^2 f}{\partial x_2 \partial x_n}(\mathbf{x}) & \cdots & \frac{\partial^2 f}{\partial x_n^2}(\mathbf{x}) \end{bmatrix}$$

That is, $H_f(\mathbf{x})$ has (i, j)-entry $f_{x_i x_j}(\mathbf{x})$, obtained by first differentiating in the x_i variable, and then in the x_j variable. Row i differentiates f_{x_i} with respect to each input variable. When the relevant mixed second partial derivatives are continuous on a ball around $\mathbf{x}$, Clairaut's Theorem implies that the Hessian is symmetric at $\mathbf{x}$.

In particular, the Hessian of a scalar-valued function of two variables can be written

$$H_f = \begin{bmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial y \partial x} \\ \frac{\partial^2 f}{\partial x \partial y} & \frac{\partial^2 f}{\partial y^2} \end{bmatrix} = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix}.$$

Activity: The Hessian of a trigonometric function (U3-LO3)

Find the Hessian matrix for the function

$$f(x, y) = \sin(x + y^2)$$

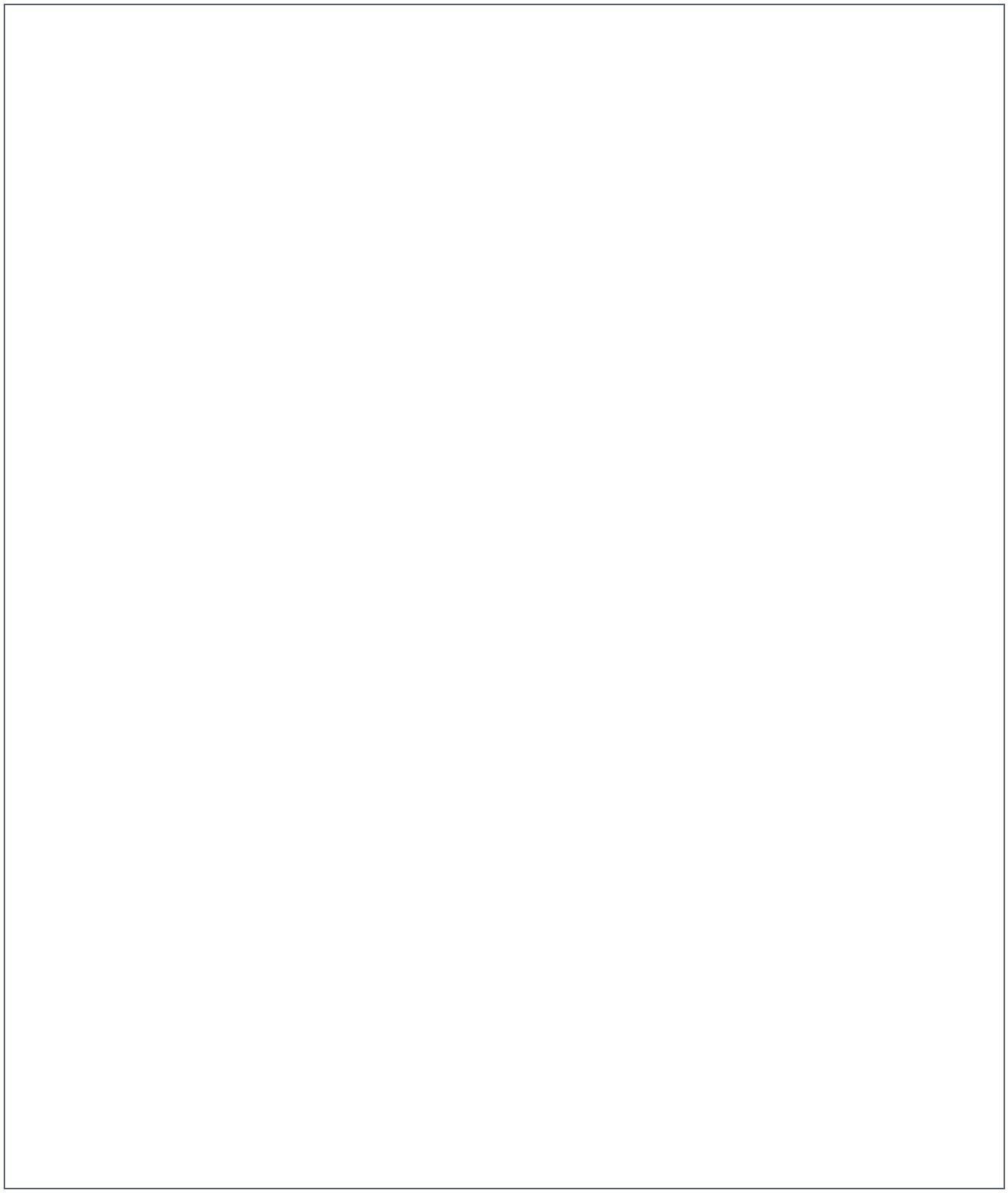
at a general point (x, y).

Activity: The Hessian of a quadratic function (U3-LO3)

Find the Hessian matrix of the function

$$f(x, y) = x^2 + xy + y^2$$

at a general point (x, y)



Activity: A gradient and Hessian in three variables (U3-LO3)

Let $f(x, y, z) = xy + yz + xz$. Compute its gradient ∇f and Hessian H_f at a general point (x, y, z).

Activity: Building a gradient and Hessian in SymPy (U3-LO3)

Collect the derivatives of $f(x, y) = 3x^2y + 2x - 3y$ into a vector and a matrix. As in Unit 2, `sp.Matrix([a, b])` creates a column matrix and `.shape` reports its numbers of rows and columns. The command `sp.hessian(f, (x, y))` constructs the Hessian with coordinates ordered as x, y.

```
import sympy as sp

x, y = sp.symbols("x y")
f = 3*x**2*y + 2*x - 3*y
grad_f = sp.Matrix([sp.diff(f, x), sp.diff(f, y)])
H = sp.hessian(f, (x, y))

print(grad_f)
print(H)
print((grad_f.shape, H.shape))
```

Output:

```
Matrix([[6*x*y + 2], [3*x**2 - 3]])
Matrix([[6*y, 6*x], [6*x, 0]])
((2, 1), (2, 2))
```

1. Write the gradient and Hessian in mathematical notation. Why is one a 2×1 column and the other a 2×2 matrix?

Activity: Building a gradient and Hessian in SymPy (U3-LO3) (continued)

2. Predict `grad_f.subs({x: 1, y: 2})` and `H.subs({x: 1, y: 2})`. Does substitution change either shape?
3. What derivative is stored in `H[0, 1]`? Explain the symmetry of `H` and why its zero bottom-right entry does not mean that f_y is identically zero.

Activity: How model weights change the loss (U3-LO3)

For the fixed data in From predictions to a scalar error function (U3-LO1, U2-LO1),

$$\mathscr{L}(w_1, w_2) = (w_1 - 1)^2 + (w_2 - 2)^2 + (w_1 + w_2 - 2)^2.$$

We will continue to use the starting weights $\mathbf{w}_0 = (1, 1)$.

1. Find both first partial derivatives and assemble $\nabla\mathscr{L}$. Evaluate it at $\mathbf{w}_0$. Interpret each component while holding the other weight fixed.
2. Compute the Hessian $H_{\mathscr{L}}$. Verify by matrix multiplication that it equals $2X^T X$ for the data matrix X.
3. The first partial derivative in w_1 vanishes at $\mathbf{w}_0$. Does this mean the loss is unchanged whenever we change w_1? Check by expanding $\mathscr{L}(1 + t, 1)$.

You can check your derivative formulas using the SymPy commands in Building a gradient and Hessian in SymPy (U3-LO3).

Jacobian matrices

For vector-valued maps, the Jacobian matrix collects the partial derivatives of all component functions.
The rows correspond to component functions. The columns describe how the output changes when one input coordinate changes.

Definition: Jacobian matrix

For a vector-valued map $F : \mathbb{R}^n \to \mathbb{R}^m$ given by

$$F(\mathbf{x}) = \begin{bmatrix} F_1(\mathbf{x}) \\ F_2(\mathbf{x}) \\ \vdots \\ F_m(\mathbf{x}) \end{bmatrix}$$

where $\mathbf{x} = (x_1, x_2, \ldots, x_n)$, the [ ] of F is the $m \times n$ matrix:

$$J_F(\mathbf{x}) = \begin{bmatrix} \frac{\partial F_1}{\partial x_1} & \frac{\partial F_1}{\partial x_2} & \cdots & \frac{\partial F_1}{\partial x_n} \\ \frac{\partial F_2}{\partial x_1} & \frac{\partial F_2}{\partial x_2} & \cdots & \frac{\partial F_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial F_m}{\partial x_1} & \frac{\partial F_m}{\partial x_2} & \cdots & \frac{\partial F_m}{\partial x_n} \end{bmatrix}$$

The Jacobian matrix generalizes the concept of the derivative to a vector-valued map of multiple variables. It provides a compact representation of all first-order partial derivatives of a function.

Activity: The Jacobian matrix of a scalar-valued function (U3-LO3)

Find the Jacobian matrix of $f(x, y, z) = xy + yz + xz$.

Activity: The Jacobian matrix of a three-component map (U3-LO3)

Compute the Jacobian matrix of $F(x, y) = (x^2 - y^2, 2xy, x + y)$.

Activity: The Jacobian matrix of a gradient (U3-LO3)

Compute the Jacobian matrix of the gradient function $\nabla f(x, y)$ for $f(x, y) = x^2 + xy + y^2$.

Note: Jacobian, gradient, and Hessian

For a scalar-valued function $f : \mathbb{R}^n \to \mathbb{R}$, the Jacobian has one row, while the gradient in Gradient is a column vector. Wherever the first partial derivatives exist,

$$J_f(\mathbf{x}) = \begin{bmatrix} \dfrac{\partial f}{\partial x_1}(\mathbf{x}) & \cdots & \dfrac{\partial f}{\partial x_n}(\mathbf{x}) \end{bmatrix} = \big(\nabla f(\mathbf{x})\big)^T.$$

Equivalently, $\nabla f(\mathbf{x}) = J_f(\mathbf{x})^T$.

If the second partial derivatives also exist, the Jacobian of the gradient is

the Hessian:

$$J_{\nabla f}(\mathbf{x}) = H_f(\mathbf{x}).$$

To see this, the ith component of ∇f is $\partial f / \partial x_i$. Row i, column j of its Jacobian therefore contains

$$\begin{aligned}\left[J_{\nabla f}(\mathbf{x})\right]_{ij} &= \frac{\partial}{\partial x_j}\left(\frac{\partial f}{\partial x_i}\right)(\mathbf{x}) \\ &= \frac{\partial^2 f}{\partial x_j \partial x_i}(\mathbf{x}) = \left[H_f(\mathbf{x})\right]_{ij},\end{aligned}$$

by the definition. Thus J_f has shape $1 \times n$, ∇f has shape $n \times 1$, and $J_{\nabla f} = H_f$ has shape $n \times n$.

Activity: Checking the Jacobian of a gradient in SymPy (U3-LO3)

The command `F.jacobian([x, y])` differentiates each component of a SymPy column matrix `F`. Rows follow the component order in `F`; columns follow the variable order `[x, y]`. Compare the scalar-valued function $f(x, y) = 3x^2y + 2x - 3y$ with its vector-valued gradient.

```
import sympy as sp

x, y = sp.symbols("x y")
f = 3*x**2*y + 2*x - 3*y
grad_f = sp.Matrix([sp.diff(f, x), sp.diff(f, y)])
J_f = sp.Matrix([f]).jacobian([x, y])
J_grad = grad_f.jacobian([x, y])
H = sp.hessian(f, (x, y))

print(J_f)
print(J_grad)
print(J_grad - H)
```

Output:

Activity: Checking the Jacobian of a gradient in SymPy (U3-LO3) (continued)

```
Matrix([[6*x*y + 2, 3*x**2 - 3]])
Matrix([[6*y, 6*x], [6*x, 0]])
Matrix([[0, 0], [0, 0]])
```

1. Why does `J_f` have one row, while `J_grad` has two? Give the input and output dimensions of each map.
2. How is `J_f` related to `grad_f`? Interpret the first row and the second column of `J_grad`.
3. What identity does the zero matrix verify? Is this a comparison at one point or a symbolic comparison for this polynomial?

